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Taebin Yoo

Sound Engineer & Emerging Researcher

Immersive Audio · Music Technology

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Education

Advanced Major Course in Audio Production | *DIMA · Mar 2026 – Feb 2027 expected*

- Bachelor's degree program; expected completion February 2027.

Associate Degree in Audio Production | *DIMA · Mar 2020 – Feb 2026*

- Dong-Ah Institute of Media and Arts · GPA 4.01 / 4.50 · Top-ranked early-admission applicant.

Research and Selected Projects

Research direction Spaces → Performers → Participation

01 SPACES

Compensation Filter for Live Venues | *2025 · Working prototype*

- Measured six speaker paths; aligned IRs and generated gain-limited FIR filters in Python for 0–40 ms and 125 Hz–4 kHz. Max/MSP applies correction; Convology XT adds the target hall's late reverberation.
- Mean spectral error fell from 2.69 to 2.24 dB at one position using the filter-design data. Other seats, new measurements, and perceptual matching remain unverified.

Live Immersive Audio Processor Studies | *2026 · Measured study*

- Measured Sound xR Image speaker gains with DME7 and Smaart, and L-ISA Studio level, frequency response, and delay through digital loopback sweeps. Compared candidate models and tested predictions on new layouts; listening and audience-seat studies remain future work.

02 PERFORMERS

Personalized Electronic Drums | *Aug 2025 · Working prototype*

- Built a Max/MSP module with manually tailored snare/ride velocity mappings, CC4 hi-hat control, and individual gain, pan, and outputs. Automatic learning and playing benefits have not been demonstrated.

Trackform Audio-to-MIDI and ML Evaluation | *2026 – Present · Working prototype*

- Integrated Transkun, Basic Pitch, ADTOF, and Demucs into a MIDI editing workflow with audition and export. ADTOF fine-tuning improved rendered validation F1 (0.695 to 0.786) but reduced F1 on 23 real recordings (0.859 to 0.816); retained pretrained weights. Correction effort and rehearsal benefits remain unevaluated.

03 PARTICIPATION

CI Interactive Ensemble Room | *2026 – Present · Working prototype*

- Built a Korean-language browser prototype with sound setup, stem mixing, cue comparisons, listening reports, and local data export using React, TypeScript, and Web Audio. Audio uses pink-noise placeholders; real recordings, device checks, and usability sessions with 2-5 CI users remain planned.

Music Production

Yamaha AFC Image | *2025 · Outdoor immersive performance*

- Designed and operated a 9.1-channel system with DME7, Dante, and L-Acoustics KARA; handled placement, alignment, routing, tuning, and live operation.

Arirang Sound Reinterpretation | *Mar – May 2026 · Dolby Atmos*

- Used Lee Doo-yong's 2003 remake as the visual source; created music with Suno AI, restored byeonsa narration with Waves ReGen, and mixed Dolby Atmos and stereo versions.

Publications Presentations and Submission Plans

Generation of Compensation Filters and a Real-Time Convolution System for Spatial Reproduction | *2026*

- 5th Electroacoustics & Stage Sound Conference, Acoustical Society of Korea. Outstanding Paper Presentation Award.

Can the Current Venue Sound Like Another Hall? Building a Spatial Reproduction System | *2026 Q3*

- Stage Sound Magazine, Vol. 19, pp. 78–87. Stage Sound Engineer Association of South Korea. Technical article in Korean.

Compensation Filter project paper | *Planned submission · IEEE VR 2027*

SA4ME: International Workshop on Spatial Audio for Multisensory Experiences

- Submission planned; not yet submitted or accepted. If accepted, the paper is expected to be published in the IEEE Xplore Digital Library.

Professional Experience

Freelance Sound Engineer | *Seoul · 2022 – Present*

- Engineered 200+ concerts, worship services, and events; operated FOH, monitors, microphones, routing, system setup, alignment, and tuning.

Church Sound Engineer | *Various churches, South Korea · 2023 – Present*

- Provided live sound across multiple churches, including one year part-time at Armed Forces Central Church, Seoul.

Technical Intern in System Design | *BLS Co., Ltd. · Jun – Jul 2025*

- Assisted with Waves eMotion LV1 demonstration preparation and technical support for live sound systems.

Academic Activities and Leadership | *DIMA SR Society · 2023 – 2024*

- Organized technical discussions and projects; served as Head of Academic Affairs in 2024.

Technical Skills

Programming and interactive audio Python · Max/MSP · Yamaha ProVisionaire
Spatial and immersive audio Dolby Atmos Renderer · Yamaha AFC Image / Sound xR Image · L-ISA · Spat Revolution · DADman
Measurement and networks Smaart · Dante Controller · Q-SYS
Production and engineering Pro Tools · Logic Pro · Live sound · Recording · Mixing · Mastering · System tuning · Audio restoration

Honors and Awards

Outstanding Paper Presentation Award · Acoustical Society of Korea · 2026
Jeong-Do Scholarship · 1st Place · SAMA Sound · 2025
Mi-Rae Scholarship · 2nd Place · SAMA Sound · 2024
Academic Excellence Scholarship · Dong-Ah Institute of Media and Arts · 2020–2026

Certifications and Training

Q-SYS Architect and Level Zero · QSC · 2026 | House of worship audio training · QSC · 2026
Smaart Operation Level 1 · Rational Acoustics · 2025 | Dante Levels 1 and 2 · Audinate · 2023
National Technical Qualification for Theatrical Arts Audio · Ministry of Culture, Sports and Tourism · 2023

Additional Experience

Korean Tutor · Preply · 2023–Present. Online Korean lessons for international learners in English.
Binaural Playback Preference Experiment · AIRIS Lab, KAIST · 2025. Participant.
Republic of Korea Army · 2021–2022. Blue House / Presidential Security Service; discharged as Sergeant.